# RIN RAY

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CRC Deformations-Geometry-Structure Postdoctoral Fellow, Universität Münster
*Eisenstein series · zeta functions and special values · the Lubin–Tate action*

## ACADEMIC EMPLOYMENT

2023–Present **CRC Postdoctoral Fellow**; Universität Münster; Arithmetic and Homotopy Theory Working Group, with Thomas Nikolaus and Christopher Deninger
- chromatic homotopy theory · Eisenstein series and the Langlands program · zeta functions and special values · the Lubin–Tate action and curves

2015 **Visiting Researcher**; Santa Fe Institute of Complex Systems; Santa Fe, NM
- multiscale physics modelling · directed type systems

2013 **Computational Chemistry Researcher**; George Mason University, VA; advised by Dimitrios Papaconstantopoulos
- superconducting transition temperature · electronic-structure calculation · augmented plane wave method · composite materials

2013 **Automated Computational Semantics & Machine Learning Researcher**; George Mason University; Fairfax, VA
- probabilistic lexicon induction · natural language processing · decipherment

2012 **Human-Computer Interaction Researcher** (Autonomous Robotics); George Washington University; Washington D.C.
- proprioceptive-only control under graceful degradation · human–robot interaction

2012 **Autonomous Aerial Vehicle Researcher**; George Mason University
- simultaneous localization and mapping (SLAM) · motion planning · visual perception for unmanned aerial vehicles in novel environments

2011 **Research Assistant in Materials Science Lab**; Mary Baldwin College; Staunton, VA
- memristive devices · resistive switching

## EDUCATION

2017–2023 **Northwestern**; PhD in Maths; advised by Paul Goerss
- *A Geometric Model of Higher K-theories at Height $h=p^{k-1}(p-1)$ via Families of Ramified Curves*

2015–2017 **UChicago**; Divisional Masters Program in Maths; advised by Peter May and Kazuya Kato
- *Calculating $\pi_*(tmf)$ at the prime 2*
- *Models of Formal Group Laws of Every Height*

2012–2013 **George Mason University**; B.S. in Computational Physics (graduated winter 2013, at 16); advised by Dimitrios Papaconstantopoulos
- *Contextual Machine Learning through the Analysis and Chunking of Partially Translated Grade 2 Braille*
- *Predicting Superconductivity Transition Temperature via the Augmented Plane Wave Model*

2010–2012 **Mary Baldwin College**; Program for the Exceptionally Gifted: Mathematics, Computer Science, and Physics; advised by John Ong

## PUBLICATIONS

2022 **Automorphisms of Abelian Varieties and Principal Polarizations**; joint with D. Lee; *Rendiconti del Circolo Matematico di Palermo*, Series 2, vol. 71, pp. 483–494; arXiv:1811.07007

2020 **Towards Directed Collapsibility**; joint with R. Belton, R. Brooks, S. Ebli, L. Fajstrup, B. T. Fasy, N. Sanderson, E. Vidaurre; *Advances in Mathematical Sciences*, vol. 21, pp. 255–271; arXiv:1902.01039

## PREPRINTS

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| Sept 2026 | **On the Chromatic Splitting Conjecture in Coheight 1**; joint with T. Barthel, L. Mann, A. Senger, T. Schlank, J. Weinstein, X. Zhou; on Jacquet–Langlands and homotopy theory |
| Sept 2025<br>V3 OCT 2026 | **Tiny and Unique: The Lubin–Tate Action at Height $p-1$ from the Minimal Curve with Its Symmetries** submitted: *AG&T*; arXiv:2509.23428 |
| July 2025 | **Towards the $p = 3$ Kervaire Invariant Problem** submitted; joint with E. Belmont; the $E_2$-page for the homotopy fixed points spectral sequence computing $\pi_*({E_6}^{hC_9})$; arXiv:2507.10157 |
| July 2025<br>V2 OCT 2026 | **Modeling Group Actions on Stacks (Especially the Lubin–Tate Action)** submitted: *Expositiones Mathematicae*; arXiv:2507.00309 |
| 2018 | **A Global Crystalline Period Map**; joint with M. Neaton and A. Pieper; from the Arizona Winter School; arXiv:1911.08615 |

## IN PROGRESS

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| Scattering | **The Unramified Eisenstein Spectrum as a Categorical Trace** manuscript; solo |
| Scattering | ***L*-genera and Localizations in *K*-theory** early stage; joint with D. Berwick-Evans and N. Pacheco-Tallaj |
| Scattering | **Habiro Characteristic Classes of *L*-function Genera** early stage; joint with F. Wagner |
| Symmetry | **Only the Gauge Grows: Moduli Stacks of *G*-Curves in Homotopy Theory at $h = p^{k-1}(p-1)$** manuscript; Part III |
| Symmetry | **The Eigenvalues of Frobenius of Artin–Schreier–Witt Curves are Gauss Sums** note; new families of Newton strata in the Torelli locus |
| Echoes | **The λ-twisted $T(h+1)$-local Zeta Function** in preparation; joint with G. Angelini-Knoll |
| Echoes | ***L*-functions and the Relative Image of *J* in Moore Spectra** conjecture, checked |
| Echoes | **Toward Categorifying Symplectic *L*-functions and Reidemeister Torsion** in preparation; joint with A. Petrov and M. Volpe |
| Echoes | **All Bernoulli Numbers in Homotopy Theory Are Shifts** early stage; joint with A. Mejia and N. Riggenbach; connecting Kervaire–Milnor to Quillen–Lichtenbaum |

## FELLOWSHIPS

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| 2022 | Nominated for **Presidential Fellowship**; Northwestern University |
| 2017 | Awarded **NSF GRFP Fellowship** (103K + 36K) |
| 2014–2016 | Awarded **Thiel Fellowship** (100K); mathematics and assistive technology; mentor Edward Frenkel |

## AWARDS

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| 2022 | **Gelfand Prize**, Northwestern University; for outstanding progress in one's own research and extraordinary contributions to the research goals of the graduate community |
| 2013 | **Outstanding Rising Senior Award** in Computational and Data Sciences, SPACS |

## SERVICE, TEACHING AND OUTREACH

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| Winter 2025 | ORGANIZER **Linearity in *p*-adic Lie Theory**; Homotopy and Arithmetic joint working group |
| Spring 2025 | ORGANIZER ***K*-Theoretic Poitou–Tate Duality**; Homotopy and Arithmetic joint working group |
| 2025 | ORGANIZER TEACHER **"Mathe trifft Kunst" (Math Meets Art)**; activity designer and co-organizer; hands-on mathematics workshops for children, Universität Münster |
| Summer 2024 | ORGANIZER **Young Topologists Meeting** |

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| 2024 | PANELIST **Queer in Math Day**; MPI Leipzig |
| Winter 2024 | ORGANIZER **Euler Characteristics and Zeta Functions**; homotopy working group |
| Winter 2023 | ORGANIZER ***K*-Theory of the Integers**; Homotopy and Arithmetic joint working group |
| 2022–2023 | FOUNDER **Mental Health Math Buddy Program** ongoing; Northwestern. Pairs first- and second-year graduate students with older graduate students, meeting once a quarter for mental health check-ups. |
| 2022 | FUNDING Established the **Fund for Work–Life Activities for Graduate Students**; Northwestern University |
| Winter 2022 | ORGANIZER **Stop Staring and Compute!**; interactive methods of computation in number theory and algebraic geometry |
| 2021–2023 | MENTOR **TGS Diversity Peer Mentorship Program**; Northwestern. Upper-year graduate students help first-year PhD students from diverse backgrounds in other subjects acclimate, providing regular check-ins, encouragement and support. |
| 2021–2023 | MENTOR **Causeway Postbaccalaureate Program**; Northwestern. Mentoring students from diverse backgrounds through a bridge programme preparing them for graduate school. |
| 2021–2023 | MENTOR TEACHER **Directed Reading Program**; Northwestern. Weekly one-on-one work with an undergraduate on a topic of their choice, ending in a talk to their peers. |
| 2020–2023 | VOLUNTEER **Evanston Animal Shelter**; fostered special-needs adult cats and raised neonatal kittens. |
| Winter 2020 | ORGANIZER **Moduli Spaces**; alternating algebraic and analytic |
| Spring 2019 | ORGANIZER **Applications of Topos Theory**; homotopy working seminar |
| Winter 2019 | ORGANIZER **TamagaWHAT**; on the proof of the Tamagawa conjecture for function fields |
| Winter 2019 | ORGANIZER **Formal Moduli Problems**; homotopy working seminar |
| Fall 2018 | ORGANIZER **Geometry for Prime Addicts**; *p*-adic geometry toward Scholze's proof of the monodromy weight conjecture |
| Fall 2018 | ORGANIZER **Factorization Homology**; homotopy working seminar |
| 2016–2017 | TEACHER **Math Circles**; weekly volunteering with children aged 6–12, South Side of Chicago |
| 2015–2017 | TEACHER MENTOR **Research Experience for Undergraduates**; University of Chicago. Summer programme with weekly lectures and interaction, guiding each undergraduate through a targeted individual research project. |
| 2014 | FOUNDING CONTRIBUTOR **Topos House** ongoing; a still-active house in South San Francisco for scientific and artistic collaboration, especially between mathematicians and biologists. |

## OTHER EMPLOYMENT

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| 2014–2016 | **Thiel Fellow** (Math and Assistive Tech Inventor); Thiel Foundation; Berkeley, CA<br>• medical and assistive technology · independent mathematics study under Edward Frenkel |
| 2013–2014 | **Mouse Vocalization Specialist & Audio Processing Engineer**; Vium; Belmont, CA<br>• computational bioacoustics · unsupervised classification of ultrasonic vocalizations (USVs) · quasi-real-time audio processing |
| 2013 | **Software Engineering Intern**; Cloudera; Palo Alto, CA<br>• Hadoop · Hive · large-scale data processing |

## SCIENTIFIC WORKS

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| 2015 | **Simplifying Multiscale Modeling**; Santa Fe Institute of Complex Systems |
| 2014 | **A New Female-Female Mouse Vocalization Discovered via Unlabeled Machine Learning**; Vium |
| 2014 | **On the Detection and Prevention of Aggression in Lab Mice via Quasi-Real Time Analysis**; Vium |

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| 2013 | **Nonprovisional patent application**; Robotic Mobility Assistive Wheelchairs, 90776.0001USP1<br>robotic mobility · pressure injury prevention |

## SELECTED INVITED TALKS

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| 2026 | **Mod *p* Langlands and the Chromatic Splitting Conjecture**; Essen Number Theory Seminar |
| 2024 | **Moduli Stacks of Curves in Homotopy Theory**; Johns Hopkins Topology Seminar |
| 2021 | **Galois Theory of Spheres**; Front Range Number Theory |
| 2020 | **Explicit Arithmetic of Jacobians**; ICERM Workshop on Arithmetic Geometry, Number Theory, and Computation |
| 2019 | **Inverse Galois Theory in the Homotopy Groups of Spheres**; Princeton CCR Colloquium |
| 2019 | **Galois Theory and Galois Representations in Homotopy Theory**; Higher School of Economics, Moscow |
| 2019 | **An Introduction to the Classic Theory of *p*-Divisible Groups**; Oberwolfach Arbeitsgemeinschaft |

## CREATIVE WORKS

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| 2026 | ***Impaction***, first play; staged in Los Angeles as part of *Nothing to See Here*, directed by Jacques Manjarrez |
| 2025 | ***Endomortis***, comic; with Petra Flurin; crayolapunk surrealist comic about grief |